

streamlit app

Phase 1 Deep Dive

Phase 2 Deep Dive

Live Predictions

Backtest Simulator



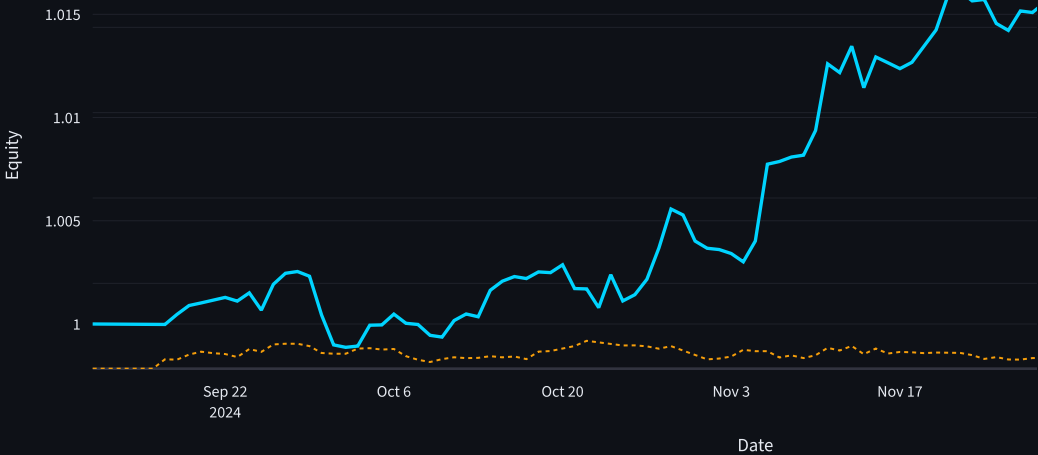
Phase 2 — Advanced Upgrades

Walk-Forward CV → Optuna HPO → Risk-Managed Backtest → SHAP Interpretability

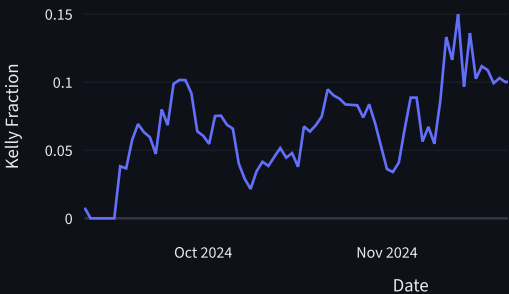


Risk-Managed Backtest — Equity Curve & Position Sizing

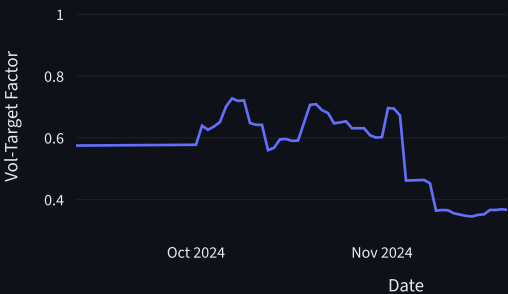
Equity Curve (Blue) + Position Size (Orange, Kelly × Vol-Target)



Kelly Fraction Sizing Over Time



Volatility Target Factor Over Time



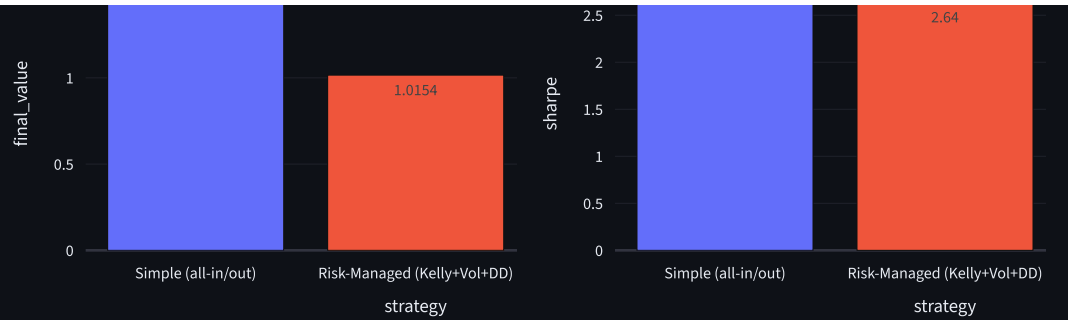
Strategy Comparison — Simple vs Risk-Managed vs Buy & Hold

Final Portfolio Value

1.5 1.5658

Sharpe Ratio

3 2.84



Full Comparison Table

strategy	final_value	sharpe	max_dd	n_tr
Simple (all-in/out)	1.5658	2.8439	-0.1206	
Risk-Managed (Kelly+Vol+DD)	1.0154	2.6369	-0.0037	
Buy & Hold	1.6156	2.9633	-0.1272	

⚙️ Optuna Hyperparameter Search Results

Best OOF Sharpe	Total Trials	Pruned Trials
2.3299	5	1 (20%)

Best Hyperparameters

Hyperparameter	Value
Learning Rate	0.000561
LSTM Units	32
Dropout	0.0
Number of Layers	1

Search Space & Walk-Forward Config

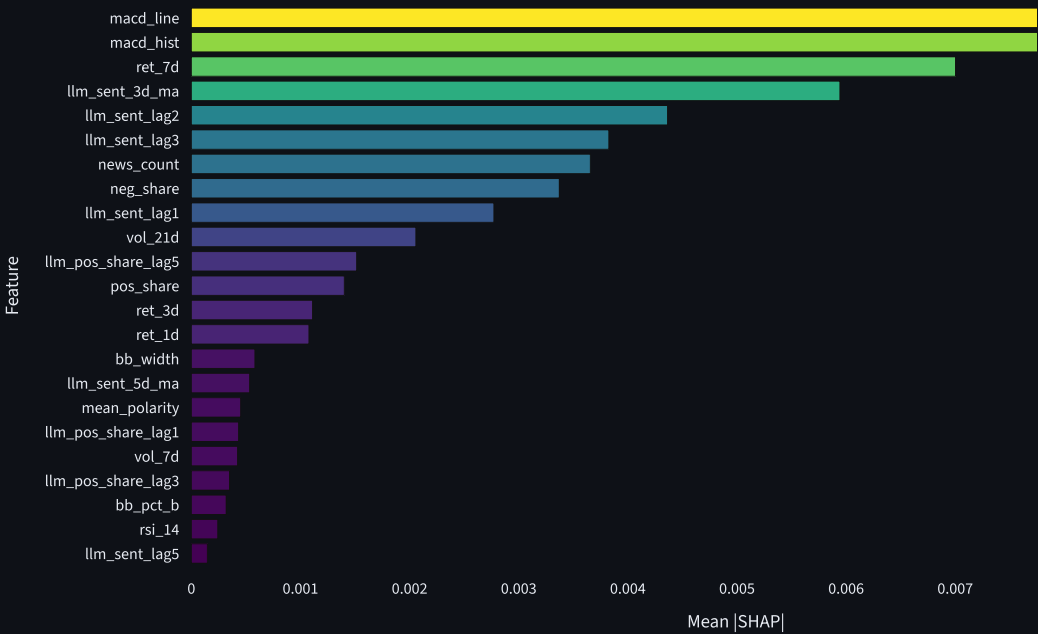
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{
  "lr": "log-uniform [1e-4, 1e-2]"
  "units": [
    0 : 32
    1 : 64
    2 : 128
  ]
  "dropout": [
    0 : 0
    1 : 0.2
    2 : 0.4
  ]
  "num_layers": [
    0 : 1
    1 : 2
  ]
}

{
  "n_folds": 5
  "min_train": 400
  "val_size": 60
}
```

}

SHAP Feature Importance

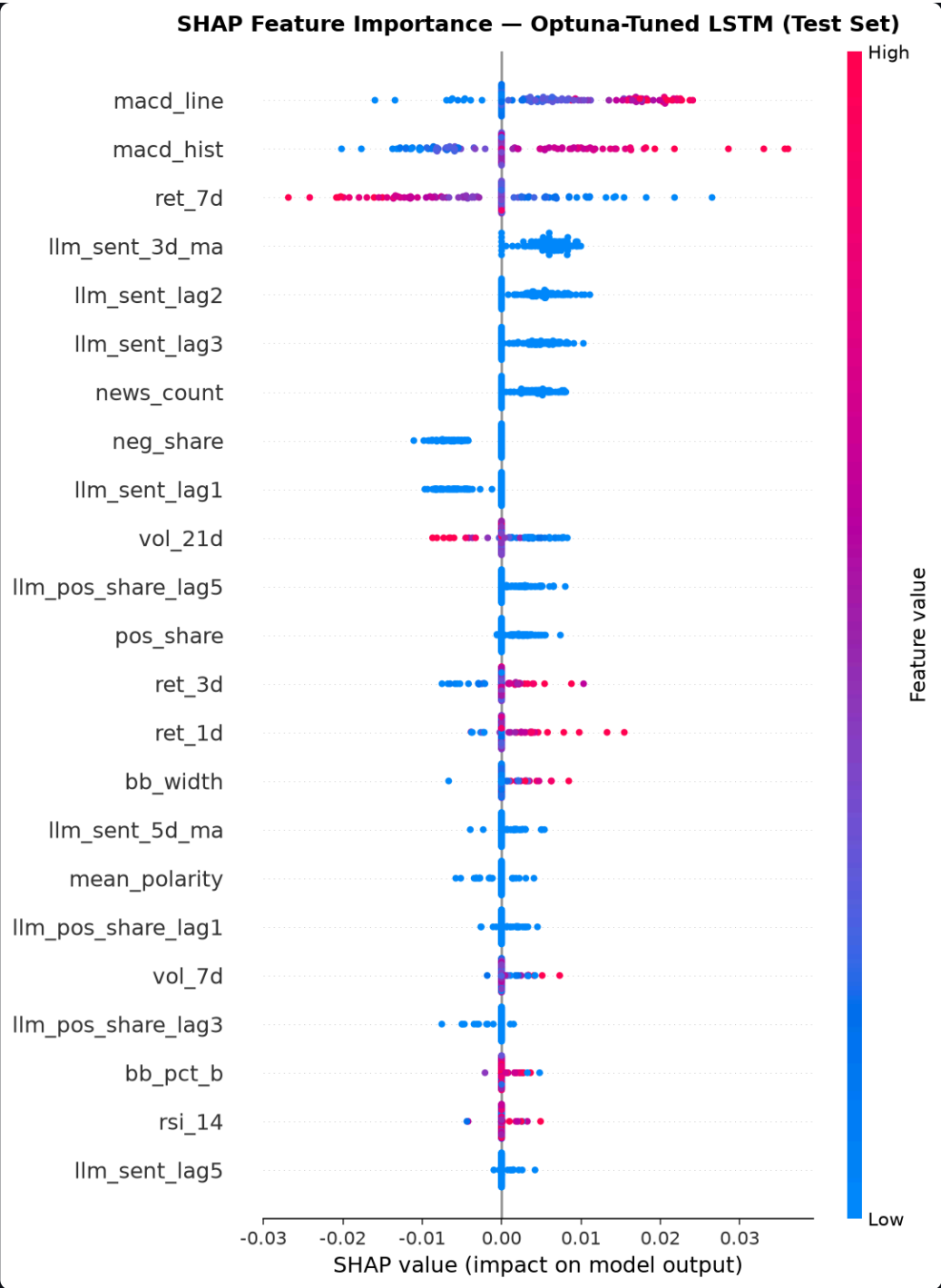
Global Feature Importance (Mean |SHAP| Value)



Top 10 Features by Mean |SHAP|

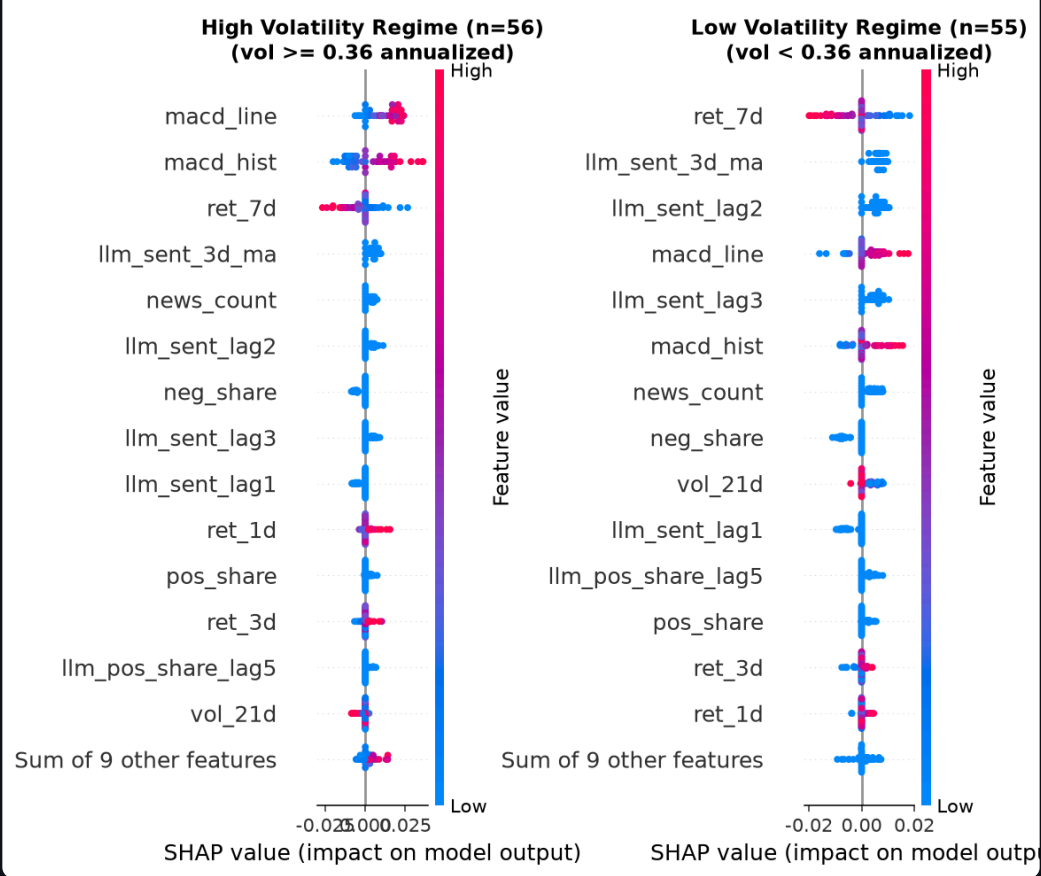
feature	mean_abs_shap
macd_line	
macd_hist	
ret_7d	
llm_sent_3d_ma	
llm_sent_lag2	
llm_sent_lag3	
news_count	
neg_share	
llm_sent_lag1	
vol_21d	

SHAP Beeswarm Summary Plot



SHAP Regime Comparison (High-Vol vs Low-Vol)

SHAP Regime Comparison: Feature Importance by Volatility



Phase 2 — Walk-forward CV, Optuna HPO, Kelly + vol-targeting + DD circuit breaker, SHAP interpretability.